

Parameters & Units

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All physical quantities in the codebase use the same unit system: **ms for time, mV for voltage, nF for capacitance, μ S for conductance, nA for current, Hz for rates.**

Time fields carry an explicit `_ms` suffix (`sim_ms`, `ref_ms_E`, `tau_gaba`); CLI flags follow the same convention (`-t-ms 600`). A Δt of 1 means 1 ms, not 1 s.

Quantities

Quantity	Unit	Typical value	Variable
Integration step	ms	0.25	<code>dt</code> , <code>DT_MS</code>
Simulation length	ms	600	<code>sim_ms</code>
Membrane time constant	ms	20 (E), 5 (I)	<code>tau_m_E</code> , <code>tau_m_I</code>
Refractory period	ms	3 (E), 1.5 (I)	<code>ref_ms_E</code> , <code>ref_ms_I</code>
AMPA decay	ms	2	<code>tau_ampa</code>
GABA decay	ms	9	<code>tau_gaba</code>
Resting / leak potential	mV	-65	<code>E_L</code>
Spike threshold	mV	-50	<code>V_th</code>
Reset potential	mV	-65	<code>V_reset</code>
AMPA reversal	mV	0	<code>E_e</code>
GABA reversal	mV	-80	<code>E_i</code>
Membrane capacitance	nF	1.0 (E), 0.5 (I)	<code>C_m_E</code> , <code>C_m_I</code>
Leak conductance	μ S	0.05 (E), 0.1 (I)	<code>g_L_E</code> , <code>g_L_I</code>
External drive	μ S	0.0006 (async), 0.003 (PING)	<code>t_e_async</code> , <code>t_e_ping</code>
Input current (CUBA)	nA	20 per spike	<code>input_scale</code>
CUBA weight std	nA	32	<code>W_STD_CUBA</code>
Max input rate	Hz	25	<code>max_rate_hz</code>
Population firing rate	Hz	20–80	<code>r_E</code> , <code>r_I</code>
Gamma frequency	Hz	30–80	<code>f_0</code>

COBA / PING biophysical constants

Used by COBA and PING. Values follow neuroscience conventions (cf. Dayan & Abbott, Gerstner *Neuronal Dynamics*); the E:I asymmetry in τ_m , C_m , g_L , τ_{ref} produces the timescale separation that makes PING dynamics possible.

Parameter	E population	I population
τ_m (ms)	20	5
C_m (nF)	1.0	0.5

g_L (μS)	0.05	0.1
τ_{ref} (ms)	3	1.5
E_L (mV)	−65	−65
V_{th} (mV)	−50	−50
V_{reset} (mV)	−65	−65
E_e (mV, reversal)	0	0
E_i (mV, reversal)	−80	−80

Synapse time constants: $\tau_{\text{AMPA}} = 2$ ms (excitation), $\tau_{\text{GABA}} = 9$ ms (inhibition). These set the ceiling on PING’s Δt -stability: once $\Delta t \gtrsim \tau_{\text{GABA}}$ the E→I→E loop cannot complete within one step.

Internal consistency

The chosen units are self-consistent — no conversion factors appear in the integration code. Two equations carry the whole system.

The membrane time constant is $\tau = C/g$. With C in nF and g in μS ,

$$\tau_{[\text{ms}]} = \frac{C_{[\text{nF}]}}{g_{[\mu\text{S}]}}$$

so $C_m = 1$ nF and $g_L = 0.05$ μS give $\tau_m = 20$ ms directly.

The LIF voltage update is $dv = (\Delta t/C)(-g_L(v - E_L) + I)$. With Δt in ms, C in nF, v, E in mV, g in μS , and I in nA,

$$dv_{[\text{mV}]} = \frac{\Delta t_{[\text{ms}]}}{C_{[\text{nF}]}} \cdot I_{[\text{nA}]}$$

because $\text{ms} \cdot \text{nA} / \text{nF} = \text{mV}$ exactly.

Conductance-current products share the same ledger: $g(v - E)$ is $\mu\text{S} \times \text{mV} = \text{nA}$, so synaptic currents fold into I alongside any direct input current without a scale factor.

Why not SI?

Pure SI (F, S, V, A, s) forces every value to a large negative exponent — $C_m = 10^{-9}$ F, $g_L = 5 \times 10^{-8}$ S, $\Delta t = 2.5 \times 10^{-4}$ s. The neuroscience convention (ms, mV, nF, μS , nA) keeps every typical value between 10^{-3} and 10^2 , which makes numerical debugging and human intuition faster. The tradeoff is that readers have to trust the unit consistency rather than verify it by plugging into SI formulas — this page is here so that trust is auditable.